

Large prime-degree isogenies

P-192 and P-224 | bounded search | 2026-10-08

Question: construct rational prime-degree maps using the new Hecke/Newton modular-polynomial and BMSS fastElkies' algorithms, including degrees above 1009.

P-192: 546 prime degrees screened; 271 split, 275 inert, 0 repeated eigenvalue. 24 explicit construction attempts; 3 passed, 21 timed out, 0 failed. 6 independently certified maps. Largest certified degree: 199.

P-224: 546 prime degrees screened; 246 split, 299 inert, 1 repeated eigenvalue. 16 explicit construction attempts; 8 passed, 8 timed out, 0 failed. 16 independently certified maps. Largest certified degree: 197.

Construction probes at or beyond degree 1009

P-192 | degree 1013 | TIMEOUT | receipt: evidence-v4/p192/ell-1013.receipt.json

P-192 | degree 2011 | TIMEOUT | receipt: evidence-v4/p192/ell-2011.receipt.json

P-192 | degree 4001 | TIMEOUT | receipt: evidence-v4/p192/ell-4001.receipt.json

P-224 | degree 1009 | TIMEOUT | receipt: evidence-v4/p224/ell-1009.receipt.json

P-224 | degree 2003 | TIMEOUT | receipt: evidence-v4/p224/ell-2003.receipt.json

P-224 | degree 4001 | TIMEOUT | receipt: evidence-v4/p224/ell-4001.receipt.json

Bounds: all primes from 67 through 4093 were screened. Every split degree through 257 was attempted, plus the first split prime at or above 509, 1009, 2003 and 4000 for each source. Each construction had a 180-second budget and a sampled 8 GiB RSS cap. Other split primes remain candidates.

Verification: the existing walker's independent implementation checks squarefreeness, torsion, subgroup closure, and the Velu codomain. Exact rational-map substitution and 20 fresh public scalar-transport checks pass for every map; each mapped nonidentity generator satisfies the published prime order.

Interpretation boundary: a split Frobenius polynomial supplies candidate eigenlines. Only completed, independently replayed kernels and maps count as certified constructions. A timeout is a resource-bounded outcome, not a nonexistence result. ECDLP-cost changes were not measured.

Sources and evidence

BMSS: <https://arxiv.org/abs/cs/0609020>

New native algorithms: isogeny_algos/docs/USAGE.md

Frozen protocol: PROTOCOL.md; raw evidence: evidence-v4/

Search: evidence-v4/search.json; independent certification: evidence-v4/replay.json

Exact target models and generators: evidence-v4/curves.json

P-192 / P-224: large prime-degree search

Explicit maps and structural screening have different verification obligations.

P-192: source model

icv1-fp192-t31607402316713927207482677199-52e4af59

EC1P192Cp192h5531c4a08bdb

urn:ec-record:1:sha256:5531c4a08bdb64b6e86a6e30e9a08a

a57edef7af15ac5f6d4d2a83a53bf2f646

6 certified maps; largest degree 199

degree 199 ->

One independently certified target

icv1-fp192-t31607402316713927207482677199-be2c363a

EC1P192Cfphec58611383c9

urn:ec-record:1:sha256:ec58611383c9eaf3570e48b9124145

7283e995dfe23c9204868a647dd2de1930

Kernel, codomain, and public subgroup transport: PASS

P-224: source model

icv1-fp224-t4733100108545601916421827343930821-fec01e99

EC1P224Cp224h8e110c585480

urn:ec-record:1:sha256:8e110c5854808bfd2283aa4b15063d

4c3052a0d742a47224e34876c6ade763bd

16 certified maps; largest degree 197

degree 197 ->

One independently certified target

icv1-fp224-t4733100108545601916421827343930821-ce849ab2

EC1P224Cfphb92e4245841c

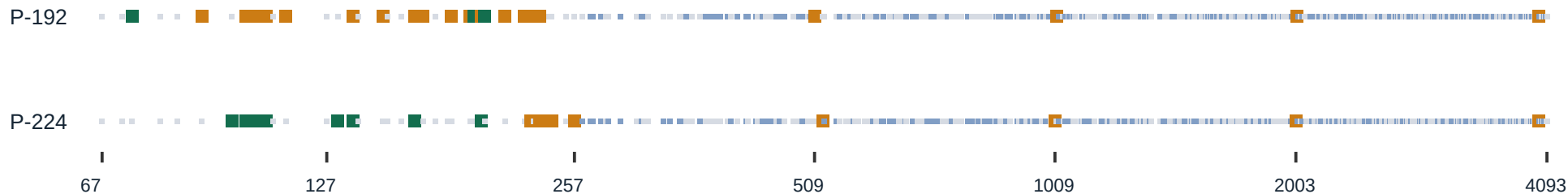
urn:ec-record:1:sha256:b92e4245841ca7a24b6f5c9566acfc

57bdc3f4c20de16eb6d5f341371fb75bdc

Kernel, codomain, and public subgroup transport: PASS

Coverage by prime degree (logarithmic degree axis)

Population: every prime in $[67, 4093]$, separately for each source. Counts are exact.



Green: replayed degree. Orange: timed out. Red: failed. Blue: split candidate only. Gray: other screen status.

Validation and reproduction

Two native fixes were required: P-192 now selects exactly three Montgomery limbs, and P-224's Hecke precondition compares the full characteristic against a small integer rather than its low word (1). The final standalone release suite passes 111 tests, including the quadratic-wrapper consistency check; the supervisor's two tests also pass.

Evidence history: evidence/ retains the initial P-192 preflight failure. evidence-v2/ retains the first 40-degree search and all 16 original P-224 assertion failures. It also preserves an interrupted P-192 degree-149 invocation without a receipt, followed by the sealed retry. Those extra invocations are separate from the final distinct-degree counts.

Final evidence-v4/ reuses the 24 sealed P-192 trials byte for byte and executes fresh P-224 trials. evidence-v3/ retains a preflight RSS-monitor shutdown race. The corrected monitor samples every 25 ms and allows at most 100 ms of unavailable samples while the process exits; persistent unavailability fails closed.

The mandated root cargo test --release --lib failed to compile with 643 pre-existing errors both before and after the rebase. Independent replay and standalone tests pass separately. The repository-wide validation and pre-push publication gate remains unresolved; see validation/lib-test-after-rebase.log.

Environment: Apple M4 Pro, 14 logical cores, 48 GiB RAM, macOS arm64. Elapsed values in process receipts are operational time-budget records on a shared host. No timing or speedup comparison is made.

Reproduce from the feature branch: build isogeny_algos in release mode; run its p192_p224_large_degree_search example with the CLI path and a NEW output directory; build this study's replay binary; replay the directory against docs/curves/registry.json; then run the report binary.

Canonical views: registered target models, native cover certificates, aliases, leaderboard rosters and browser identities are refreshed from the replay. Existing IC measurement bytes, the boundary ledger and progress timeline retain their performance results. See RESULTS.md for validation and delivery status.

Dependent traits catalogue: its native exporter cannot build because the unchanged root library has 485 compiler errors. The old 121-row traits file remains unchanged and stale. Its parser also lacks existing Montgomery and Edwards model forms. These unresolved gates are recorded explicitly, without claiming a completed traits refresh.

Source: report.rs. Final progress: SEARCH-final.log. MANIFEST.json binds protocol, source, executable versions and evidence bytes. SEARCH.svg is editable. Failures and timeouts remain alongside the completed maps.